In [1]:
```
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.preprocessing import MinMaxScaler,LabelEncoder
import joblib
```

In [2]:
```
def reduce_mem_usage(df):
    start_mem = df.memory_usage().sum() / 1024**2
    print('Memory usage of dataframe is {:.2f} MB'.format(start_mem))

    for col in df.columns:
        col_type = str(df[col].dtype)

        if col_type == 'object' or col_type == 'category':
            df[col] = df[col].astype('category')
            continue

        try:
            c_min = df[col].min()
            c_max = df[col].max()
        except:
            df[col] = df[col].astype('category')
            continue

        if 'int' in col_type:
            if c_min > np.iinfo(np.int8).min and c_max < np.iinfo(np.int8).m
                df[col] = df[col].astype(np.int8)
            elif c_min > np.iinfo(np.int16).min and c_max < np.iinfo(np.int1
                df[col] = df[col].astype(np.int16)
            elif c_min > np.iinfo(np.int32).min and c_max < np.iinfo(np.int3
                df[col] = df[col].astype(np.int32)
            elif c_min > np.iinfo(np.int64).min and c_max < np.iinfo(np.int6
                df[col] = df[col].astype(np.int64)
        elif 'float' in col_type:
            if c_min > np.finfo(np.float16).min and c_max < np.finfo(np.floa
                df[col] = df[col].astype(np.float16)
            elif c_min > np.finfo(np.float32).min and c_max < np.finfo(np.fl
                df[col] = df[col].astype(np.float32)
            else:
                df[col] = df[col].astype(np.float64)
        else:
            df[col] = df[col].astype('category')

    end_mem = df.memory_usage().sum() / 1024**2
    print('Memory usage after optimization is: {:.2f} MB'.format(end_mem))
    print('Decreased by {:.1f}%'.format(100 * (start_mem - end_mem) / start_

    return df
```

In [3]:
```
df_id = pd.read_csv("/home/mostafashazly/Desktop/ieee-fraud-detection/ieee-f
df_id.head()
```

Out[3]:

	TransactionID	id_01	id_02	id_03	id_04	id_05	id_06	id_07	id_08	id_09	...	
0	2987004	0.0	70787.0	NaN	NaN	NaN	NaN	NaN	NaN	NaN	...	sam bro
1	2987008	-5.0	98945.0	NaN	NaN	0.0	-5.0	NaN	NaN	NaN	...	m
2	2987010	-5.0	191631.0	0.0	0.0	0.0	0.0	NaN	NaN	0.0	...	ch
3	2987011	-5.0	221832.0	NaN	NaN	0.0	-6.0	NaN	NaN	NaN	...	ch
4	2987016	0.0	7460.0	0.0	0.0	1.0	0.0	NaN	NaN	0.0	...	ch

5 rows × 41 columns

In [4]:
```
df_tran = pd.read_csv("/home/mostafashazly/Desktop/ieee-fraud-detection/ieee
df_tran.head()
```

Out[4]:

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	card3
0	2987000	0	86400	68.5	W	13926	NaN	150.0
1	2987001	0	86401	29.0	W	2755	404.0	150.0
2	2987002	0	86469	59.0	W	4663	490.0	150.0
3	2987003	0	86499	50.0	W	18132	567.0	150.0
4	2987004	0	86506	50.0	H	4497	514.0	150.0

5 rows × 394 columns

In [5]:
```
df_trainid = reduce_mem_usage(df_id)
df_trans = reduce_mem_usage(df_tran)
```

```
Memory usage of dataframe is 56.51 MB
Memory usage after optimization is: 9.95 MB
Decreased by 82.4%
Memory usage of dataframe is 1791.67 MB
Memory usage after optimization is: 487.16 MB
Decreased by 72.8%
```

In [6]:
```
list_tittle = df_trainid[['TransactionID','DeviceType','DeviceInfo']]
final_data_for_train = pd.merge(df_trans,list_tittle,on="TransactionID",how=
```

In [7]:
```
final_data_for_train['TransactionAmt'] = final_data_for_train['TransactionAm
final_data_for_train['TransactionAmt'].nlargest(10)
```

Out[7]:
```
274336    31936.0
274339    31936.0
296021     6452.0
248413     6084.0
384603     5544.0
275529     5420.0
275535     5420.0
584767     5368.0
171451     5280.0
303106     5280.0
Name: TransactionAmt, dtype: float32
```

In [8]:
```
final_data_for_train.head(10)
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[8]:

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	card3
0	2987000	0	86400	68.5	W	13926	NaN	150.0
1	2987001	0	86401	29.0	W	2755	404.0	150.0
2	2987002	0	86469	59.0	W	4663	490.0	150.0
3	2987003	0	86499	50.0	W	18132	567.0	150.0
4	2987004	0	86506	50.0	H	4497	514.0	150.0
5	2987005	0	86510	49.0	W	5937	555.0	150.0
6	2987006	0	86522	159.0	W	12308	360.0	150.0
7	2987007	0	86529	422.5	W	12695	490.0	150.0
8	2987008	0	86535	15.0	H	2803	100.0	150.0
9	2987009	0	86536	117.0	W	17399	111.0	150.0

10 rows × 396 columns

In [9]:
```
fraud_data = final_data_for_train[final_data_for_train['isFraud'] == 1]
fraud_data.info()
print('\nvalues',fraud_data['TransactionAmt'].mean(),'\n')
fraud_data["TransactionAmt"].value_counts(normalize=True)
```

```
<class 'pandas.DataFrame'>
Index: 20663 entries, 203 to 590526
Columns: 396 entries, TransactionID to DeviceInfo
dtypes: category(16), float16(331), float32(45), int16(1), int32(2), int8(1)
memory usage: 17.3 MB

values 149.24435
```

Out[9]:
```
TransactionAmt
117.000000     0.034796
59.000000      0.031264
150.000000     0.027392
100.000000     0.025456
49.000000      0.022698
                 ...
1220.000000    0.000048
4.628906       0.000048
148.000000     0.000048
24.656250      0.000048
2.890625       0.000048
Name: proportion, Length: 2126, dtype: float64
```

In [10]:
```
number_of_duplicats = fraud_data.duplicated().sum()
number_of_missing_values = fraud_data.isna().sum()
print(f"number_of_duplicats is: {number_of_duplicats}\n")
print(f"number_of_missing_values is: {number_of_missing_values}")
```

```
number_of_duplicats is: 0

number_of_missing_values is: TransactionID         0
isFraud               0
TransactionDT         0
TransactionAmt        0
ProductCD             0
                  ...
V337              16977
V338              16977
V339              16977
DeviceType         9452
DeviceInfo        12056
Length: 396, dtype: int64
```

In [11]:
```
drop_col = fraud_data.columns[(fraud_data.isna().mean() >=0.8)].to_list()
```

In [12]:
```
data_droped = fraud_data.drop(columns=drop_col)
data_droped.head(10)
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[12]:

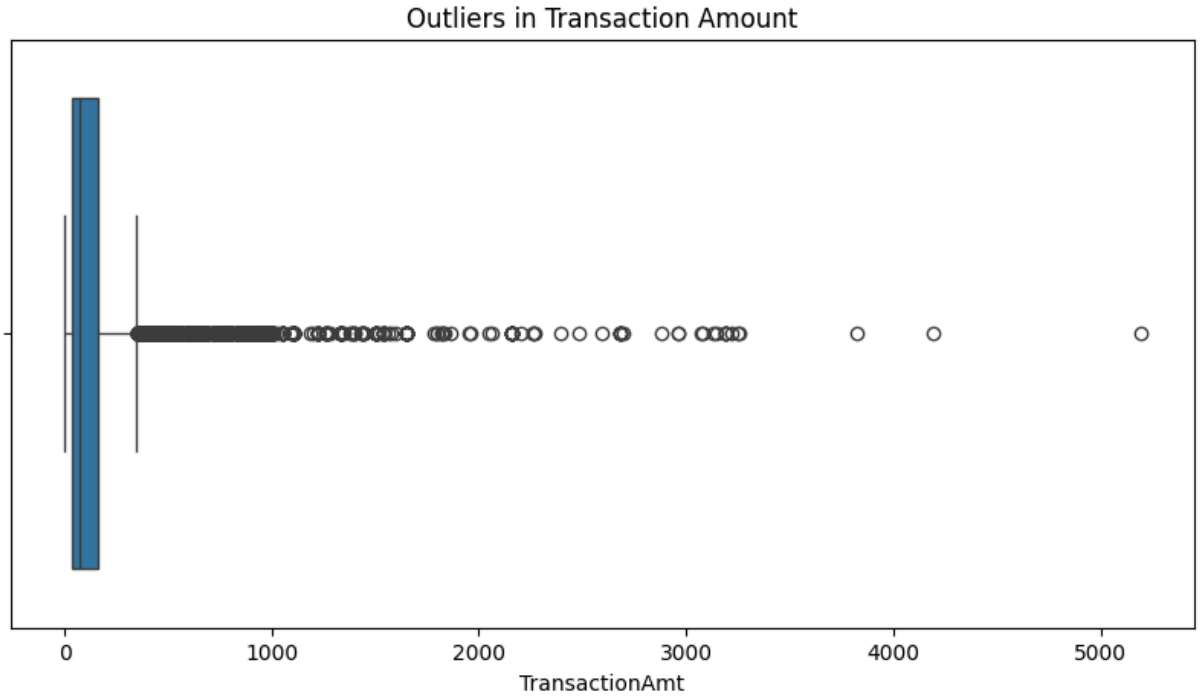
	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	ca
203	2987203	1	89760	445.000000	W	18268	583.0	15
240	2987240	1	90193	37.093750	C	13413	103.0	18
243	2987243	1	90246	37.093750	C	13413	103.0	18
245	2987245	1	90295	37.093750	C	13413	103.0	18
288	2987288	1	90986	155.500000	C	16578	545.0	18
367	2987367	1	92350	225.000000	R	4425	562.0	15
405	2987405	1	92999	90.562500	C	4504	500.0	18
630	2987630	1	97843	12.328125	C	5812	408.0	18
683	2987683	1	99584	124.375000	C	5812	408.0	18
736	2987736	1	100591	100.000000	W	15063	NaN	15

10 rows × 348 columns

In [13]:
```
print(data_droped['DeviceType'].isna().sum())
print(data_droped['DeviceInfo'].isna().sum())
```

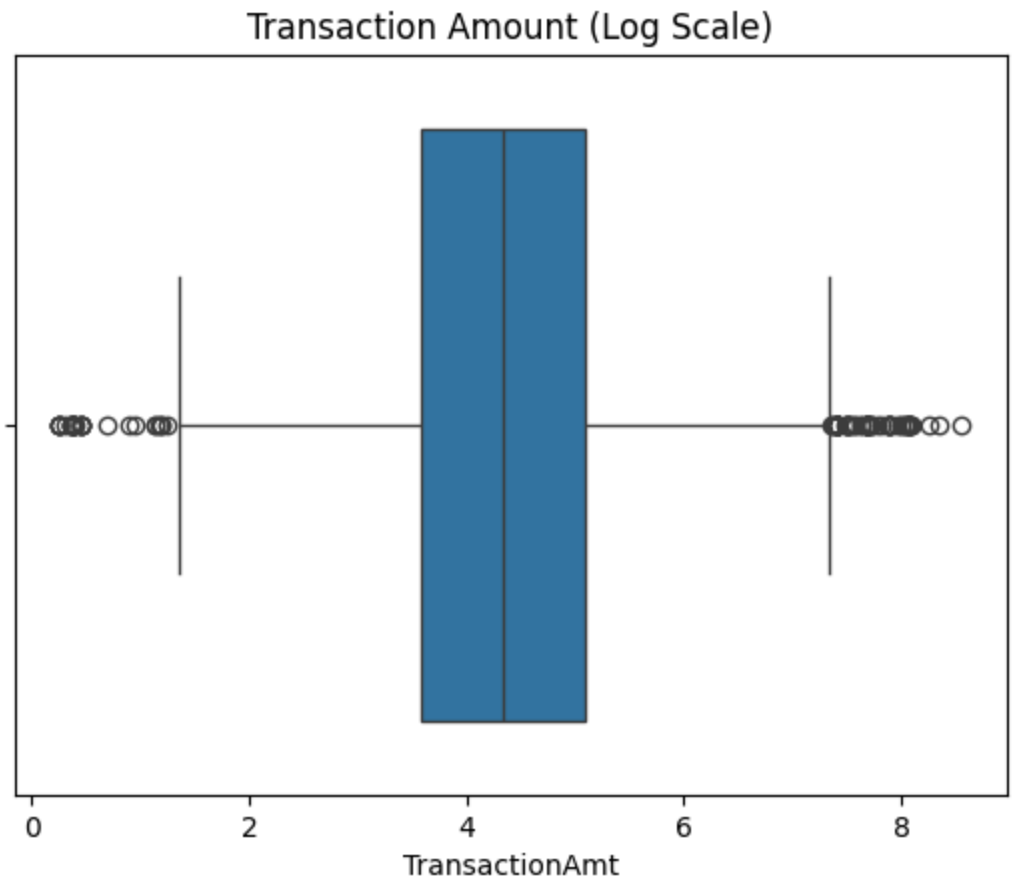
```
9452
12056
```

In [14]:
```
plt.figure(figsize=(10, 5))
sns.boxplot(data=data_droped, x='TransactionAmt')
plt.title('Outliers in Transaction Amount')
plt.show()
```



In [15]:
```
sns.boxplot(data=data_droped, x=np.log1p(data_droped['TransactionAmt']))
plt.title('Transaction Amount (Log Scale)')
```

Out[15]:
```
Text(0.5, 1.0, 'Transaction Amount (Log Scale)')
```

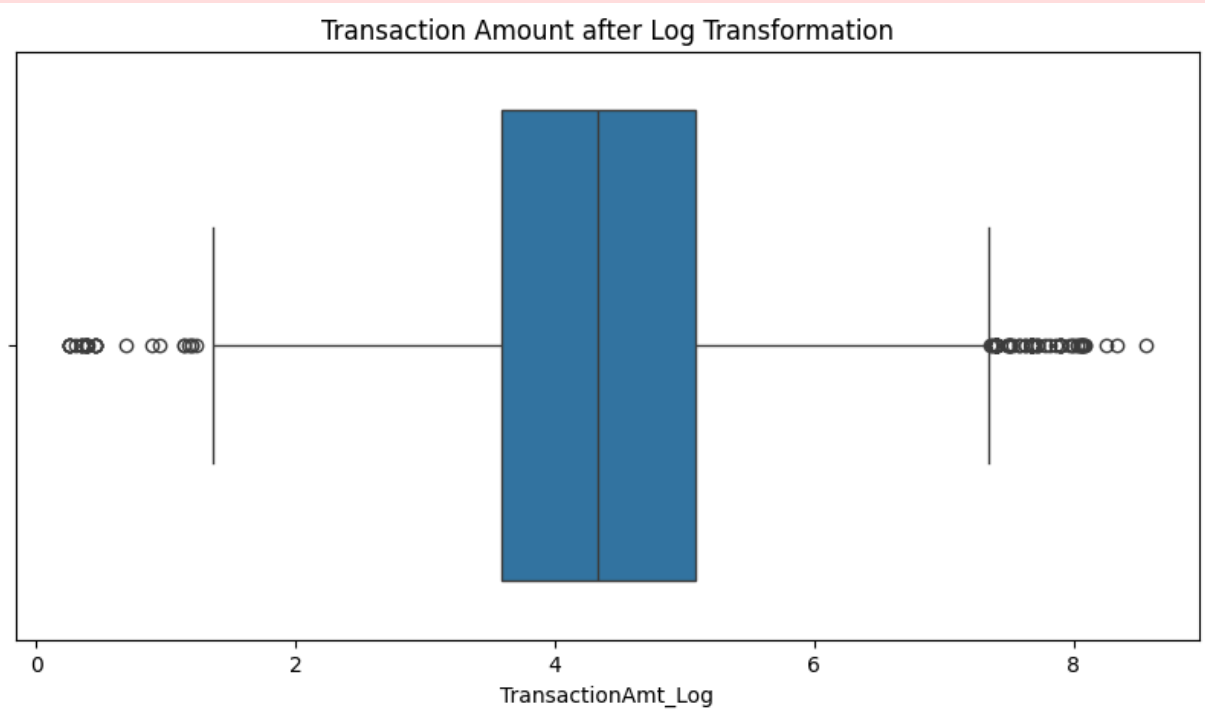


In [16]:
```
data_droped['TransactionAmt'].nlargest(15)
```

Out[16]:
```
422708    5192.0
252544    4192.0
468844    3822.0
449956    3260.0
226827    3248.0
296902    3220.0
369285    3192.0
230736    3190.0
369349    3190.0
434871    3190.0
203082    3148.0
563264    3134.0
26694     3082.0
504411    3076.0
431094    2964.0
Name: TransactionAmt, dtype: float32
```

In [17]:
```
data_droped['TransactionAmt_Log'] = np.log1p(data_droped['TransactionAmt'])

plt.figure(figsize=(10, 5))
sns.boxplot(data=data_droped, x='TransactionAmt_Log')
plt.title('Transaction Amount after Log Transformation')
plt.show()
```

```
/tmp/ipykernel_9564/1268425771.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  data_droped['TransactionAmt_Log'] = np.log1p(data_droped['TransactionAm
t'])
```



In [18]:
```
data_droped['TransactionAmt_Log'].nlargest()
```

Out[18]:
```
422708    8.555067
252544    8.341172
468844    8.248791
449956    8.089789
226827    8.086102
Name: TransactionAmt_Log, dtype: float32
```

In [19]:
```
data_os = data_droped[(data_droped['DeviceInfo'].isna()) & (data_droped['Dev
print(data_os['TransactionAmt_Log'].nlargest())
```

```
422708    8.555067
252544    8.341172
468844    8.248791
449956    8.089789
226827    8.086102
Name: TransactionAmt_Log, dtype: float32
```

In [20]:
```
data_os = data_droped[(data_droped['DeviceInfo'].notna()) & (data_droped['De
print(data_os['TransactionAmt_Log'].nlargest())
```

```
533442    7.496098
227775    7.313887
265557    7.313887
414035    7.313887
414104    7.313887
Name: TransactionAmt_Log, dtype: float32
```

In [21]:
```
data_droped['DeviceInfo'] = data_droped['DeviceInfo'].cat.add_categories('Ur
data_droped['DeviceType'] = data_droped['DeviceType'].cat.add_categories('Ur


data_droped['DeviceInfo'] = data_droped['DeviceInfo'].fillna('Unknown')
data_droped['DeviceType'] = data_droped['DeviceType'].fillna('Unknown')

print(data_droped.groupby('DeviceType')['TransactionAmt'].mean())
```

```
DeviceType
desktop     80.999855
mobile      96.592117
Unknown    220.857086
Name: TransactionAmt, dtype: float32
```

In [22]:
```
data_droped.head(10)
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[22]:

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	ca
203	2987203	1	89760	445.000000	W	18268	583.0	15
240	2987240	1	90193	37.093750	C	13413	103.0	18
243	2987243	1	90246	37.093750	C	13413	103.0	18
245	2987245	1	90295	37.093750	C	13413	103.0	18
288	2987288	1	90986	155.500000	C	16578	545.0	18
367	2987367	1	92350	225.000000	R	4425	562.0	15
405	2987405	1	92999	90.562500	C	4504	500.0	18
630	2987630	1	97843	12.328125	C	5812	408.0	18
683	2987683	1	99584	124.375000	C	5812	408.0	18
736	2987736	1	100591	100.000000	W	15063	NaN	15

10 rows × 349 columns

In [23]:
```
cat_cols = data_droped.select_dtypes(include=['category']).columns.tolist()
print(cat_cols)
```

```
['ProductCD', 'card4', 'card6', 'P_emaildomain', 'R_emaildomain', 'M1', 'M
2', 'M3', 'M4', 'M5', 'M6', 'M7', 'M8', 'M9', 'DeviceType', 'DeviceInfo']
```

In [24]:
```
print(data_droped['card1'].isna().sum())
print(data_droped['card2'].isna().sum())
print(data_droped['card3'].isna().sum())
data_droped['card2']=data_droped['card2'].astype('Float32')

print('\n',data_droped.groupby('card1')['TransactionAmt'].nlargest())
```

```
0
423
39

 card1
1015   468404    450.0000
1016   294990    100.0000
1022   178194    300.0000
1030   564074    125.0000
1039   472939     77.9375
                    ...
18375  276403    365.0000
       423272    350.5000
       194457    335.0000
       233556    300.0000
       236123    300.0000
Name: TransactionAmt, Length: 4757, dtype: float32
```

In [25]:
```
print(data_droped.groupby('card2')['TransactionAmt'].nlargest())
```

```
card2
100.0  422990    554.0000
       257199    450.0000
       307230    450.0000
       307231    450.0000
       318653    450.0000
                   ...
599.0  576858     75.0000
       277006     65.0000
600.0  572340     59.0000
       553737     49.0000
       581181     47.9375
Name: TransactionAmt, Length: 1187, dtype: float32
```

In [26]:
```
data_droped['card3'] = data_droped['card3'].astype('Float32')
print(data_droped.groupby('card3')['TransactionAmt'].nlargest())
```

```
card3
100.0  556051    132.7500
       556057    132.7500
       431609     86.3125
       379991     85.0625
       380008     85.0625
                   ...
223.0  297661    300.0000
       234774    200.0000
224.0  376847     27.1875
229.0  193129    664.0000
231.0  568160    150.0000
Name: TransactionAmt, Length: 181, dtype: float32
```

In [27]:
```
card_info = data_droped[data_droped['card1'] == 18375]

print(card_info[['TransactionAmt','DeviceType', 'DeviceInfo', 'card1', 'carc
```

```
        TransactionAmt DeviceType DeviceInfo  card1  card2  card3 card4
194457           335.0    Unknown    Unknown  18375  523.0  150.0  visa
206617            30.0    desktop    Windows  18375  174.0  150.0  visa
218662            25.0    desktop    Windows  18375  174.0  150.0  visa
226111            30.0    desktop    Windows  18375  174.0  150.0  visa
226121            30.0    desktop    Windows  18375  174.0  150.0  visa
```

In [28]:
```
card_info = data_droped[data_droped['card2'] == 100.0]

print(card_info[['TransactionAmt','DeviceType', 'DeviceInfo', 'card1', 'carc
```

```
       TransactionAmt DeviceType DeviceInfo  card1  card2  card3 card4
1643            110.0    Unknown    Unknown   2803  100.0  150.0  visa
7462            265.0    Unknown    Unknown  11106  100.0  150.0  visa
13601            75.0    Unknown    Unknown   2803  100.0  150.0  visa
13633            83.5    Unknown    Unknown   2803  100.0  150.0  visa
15957           226.0    Unknown    Unknown   2803  100.0  150.0  visa
```

In [29]:
```
card_info = data_droped[data_droped['card3'] == 100.0]
```

```
print(card_info[['TransactionAmt','DeviceType', 'DeviceInfo', 'card1', 'carc
```

```
        TransactionAmt DeviceType DeviceInfo  card1  card2  card3       card
4
226915        8.460938    desktop    Unknown  13822  555.0  100.0  mastercar
d
379991       85.062500    desktop    Windows  17392  555.0  100.0  mastercar
d
380008       85.062500    desktop    Windows  17392  555.0  100.0  mastercar
d
380046       85.062500    desktop    Windows  17392  555.0  100.0  mastercar
d
390249       54.937500    desktop    Unknown  17532  555.0  100.0  mastercar
d
```

In [30]:
```
data_droped.head(10)
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[30]:

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	ca
203	2987203	1	89760	445.000000	W	18268	583.0	15
240	2987240	1	90193	37.093750	C	13413	103.0	18
243	2987243	1	90246	37.093750	C	13413	103.0	18
245	2987245	1	90295	37.093750	C	13413	103.0	18
288	2987288	1	90986	155.500000	C	16578	545.0	18
367	2987367	1	92350	225.000000	R	4425	562.0	15
405	2987405	1	92999	90.562500	C	4504	500.0	18
630	2987630	1	97843	12.328125	C	5812	408.0	18
683	2987683	1	99584	124.375000	C	5812	408.0	18
736	2987736	1	100591	100.000000	W	15063	<NA>	15

10 rows × 349 columns

In [31]:
```
categorical_cols = data_droped.select_dtypes(include=['object', 'category'])

numerical_cols = data_droped.select_dtypes(include=['number']).columns.tolis

print("Categorical Columns:", categorical_cols)
print("\nNumerical Columns:", numerical_cols)
```

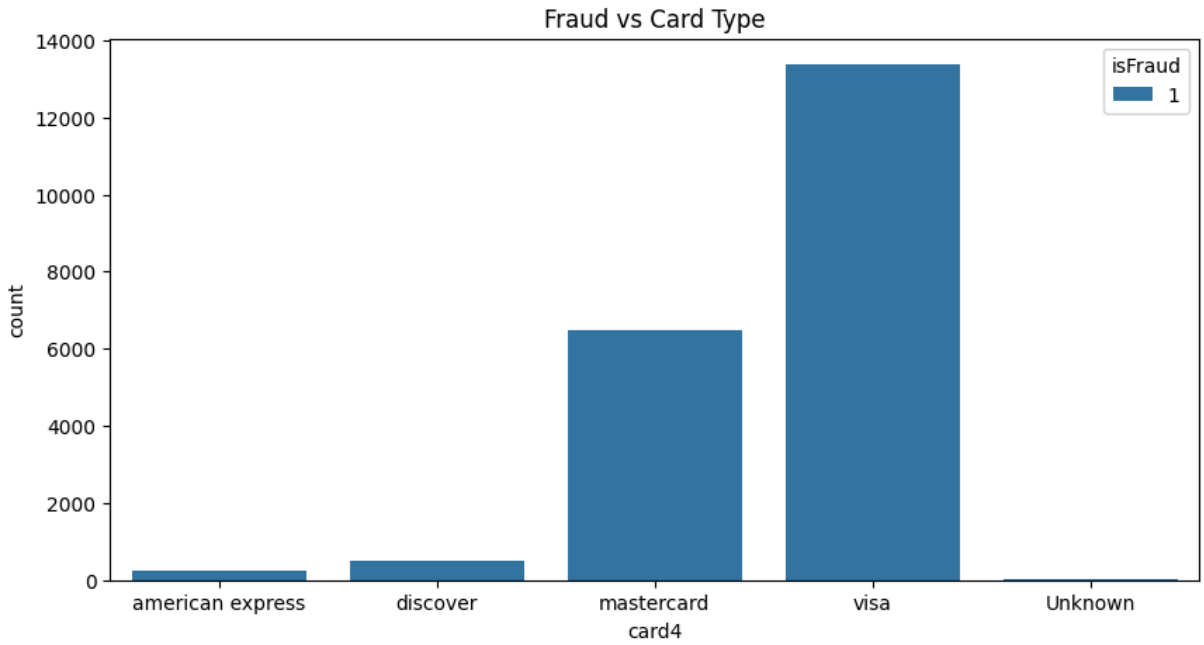
```
Categorical Columns: ['ProductCD', 'card4', 'card6', 'P_emaildomain', 'R_ema
ildomain', 'M1', 'M2', 'M3', 'M4', 'M5', 'M6', 'M7', 'M8', 'M9', 'DeviceTyp
e', 'DeviceInfo']

Numerical Columns: ['TransactionID', 'isFraud', 'TransactionDT', 'Transactio
nAmt', 'card1', 'card2', 'card3', 'card5', 'addr1', 'addr2', 'dist1', 'C1',
'C2', 'C3', 'C4', 'C5', 'C6', 'C7', 'C8', 'C9', 'C10', 'C11', 'C12', 'C13',
'C14', 'D1', 'D2', 'D3', 'D4', 'D5', 'D6', 'D7', 'D8', 'D9', 'D10', 'D11',
'D12', 'D13', 'D14', 'D15', 'V1', 'V2', 'V3', 'V4', 'V5', 'V6', 'V7', 'V8',
'V9', 'V10', 'V11', 'V12', 'V13', 'V14', 'V15', 'V16', 'V17', 'V18', 'V19',
'V20', 'V21', 'V22', 'V23', 'V24', 'V25', 'V26', 'V27', 'V28', 'V29', 'V30',
'V31', 'V32', 'V33', 'V34', 'V35', 'V36', 'V37', 'V38', 'V39', 'V40', 'V41',
'V42', 'V43', 'V44', 'V45', 'V46', 'V47', 'V48', 'V49', 'V50', 'V51', 'V52',
'V53', 'V54', 'V55', 'V56', 'V57', 'V58', 'V59', 'V60', 'V61', 'V62', 'V63',
'V64', 'V65', 'V66', 'V67', 'V68', 'V69', 'V70', 'V71', 'V72', 'V73', 'V74',
'V75', 'V76', 'V77', 'V78', 'V79', 'V80', 'V81', 'V82', 'V83', 'V84', 'V85',
'V86', 'V87', 'V88', 'V89', 'V90', 'V91', 'V92', 'V93', 'V94', 'V95', 'V96',
'V97', 'V98', 'V99', 'V100', 'V101', 'V102', 'V103', 'V104', 'V105', 'V106',
'V107', 'V108', 'V109', 'V110', 'V111', 'V112', 'V113', 'V114', 'V115', 'V11
6', 'V117', 'V118', 'V119', 'V120', 'V121', 'V122', 'V123', 'V124', 'V125',
'V126', 'V127', 'V128', 'V129', 'V130', 'V131', 'V132', 'V133', 'V134', 'V13
5', 'V136', 'V137', 'V167', 'V168', 'V169', 'V170', 'V171', 'V172', 'V173',
'V174', 'V175', 'V176', 'V177', 'V178', 'V179', 'V180', 'V181', 'V182', 'V18
3', 'V184', 'V185', 'V186', 'V187', 'V188', 'V189', 'V190', 'V191', 'V192',
'V193', 'V194', 'V195', 'V196', 'V197', 'V198', 'V199', 'V200', 'V201', 'V20
2', 'V203', 'V204', 'V205', 'V206', 'V207', 'V208', 'V209', 'V210', 'V211',
'V212', 'V213', 'V214', 'V215', 'V216', 'V217', 'V218', 'V219', 'V220', 'V22
1', 'V222', 'V223', 'V224', 'V225', 'V226', 'V227', 'V228', 'V229', 'V230',
'V231', 'V232', 'V233', 'V234', 'V235', 'V236', 'V237', 'V238', 'V239', 'V24
0', 'V241', 'V242', 'V243', 'V244', 'V245', 'V246', 'V247', 'V248', 'V249',
'V250', 'V251', 'V252', 'V253', 'V254', 'V255', 'V256', 'V257', 'V258', 'V25
9', 'V260', 'V261', 'V262', 'V263', 'V264', 'V265', 'V266', 'V267', 'V268',
'V269', 'V270', 'V271', 'V272', 'V273', 'V274', 'V275', 'V276', 'V277', 'V27
8', 'V279', 'V280', 'V281', 'V282', 'V283', 'V284', 'V285', 'V286', 'V287',
'V288', 'V289', 'V290', 'V291', 'V292', 'V293', 'V294', 'V295', 'V296', 'V29
7', 'V298', 'V299', 'V300', 'V301', 'V302', 'V303', 'V304', 'V305', 'V306',
'V307', 'V308', 'V309', 'V310', 'V311', 'V312', 'V313', 'V314', 'V315', 'V31
6', 'V317', 'V318', 'V319', 'V320', 'V321', 'TransactionAmt_Log']
```

In [32]:
```
for col in categorical_cols:
    if data_droped[col].dtype.name == 'category':
        if 'Unknown' not in data_droped[col].cat.categories:
            data_droped[col] = data_droped[col].cat.add_categories('Unknown'

    data_droped[col] = data_droped[col].fillna('Unknown')

data_droped[numerical_cols] = data_droped[numerical_cols].fillna(-1)
```

In [33]:
```
print("Total Missing Values:", data_droped.isna().sum().sum())
```

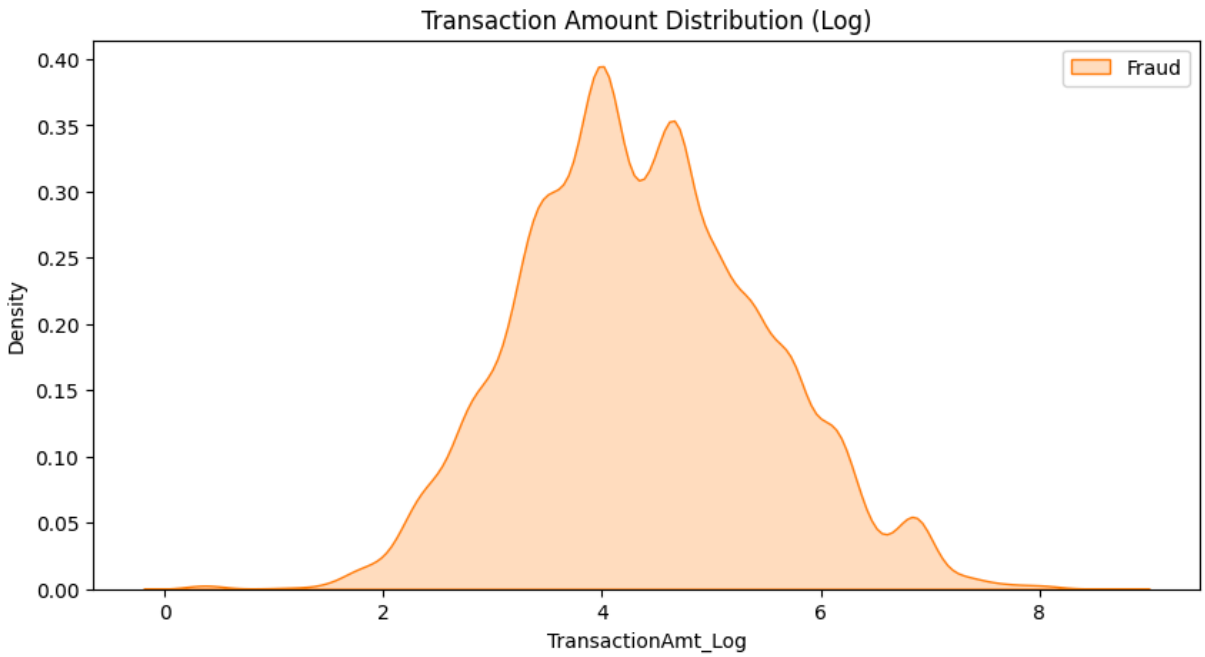
```
Total Missing Values: 0
```

In [34]:
```
plt.figure(figsize=(10,5))
sns.countplot(data=data_droped, x='card4', hue='isFraud')
plt.title('Fraud vs Card Type')
plt.show()
```



In [35]:
```
plt.figure(figsize=(10,5))
sns.kdeplot(data=data_droped[data_droped['isFraud']==0]['TransactionAmt_Log'
sns.kdeplot(data=data_droped[data_droped['isFraud']==1]['TransactionAmt_Log'
plt.title('Transaction Amount Distribution (Log)')
plt.legend()
plt.show()
```

```
/tmp/ipykernel_9564/4021991613.py:2: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=data_droped[data_droped['isFraud']==0]['TransactionAmt_Lo
g'], label='Normal', shade=True)
/tmp/ipykernel_9564/4021991613.py:3: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=data_droped[data_droped['isFraud']==1]['TransactionAmt_Lo
g'], label='Fraud', shade=True)
```



In [36]:
```
drop_col_all = final_data_for_train.columns[(final_data_for_train.isna().mea
df_clean = final_data_for_train.drop(columns=drop_col_all)
```

In [37]:
```
df_clean['TransactionAmt_Log'] = np.log1p(df_clean['TransactionAmt'])
```

```
/tmp/ipykernel_9564/2481387589.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['TransactionAmt_Log'] = np.log1p(df_clean['TransactionAmt'])
```

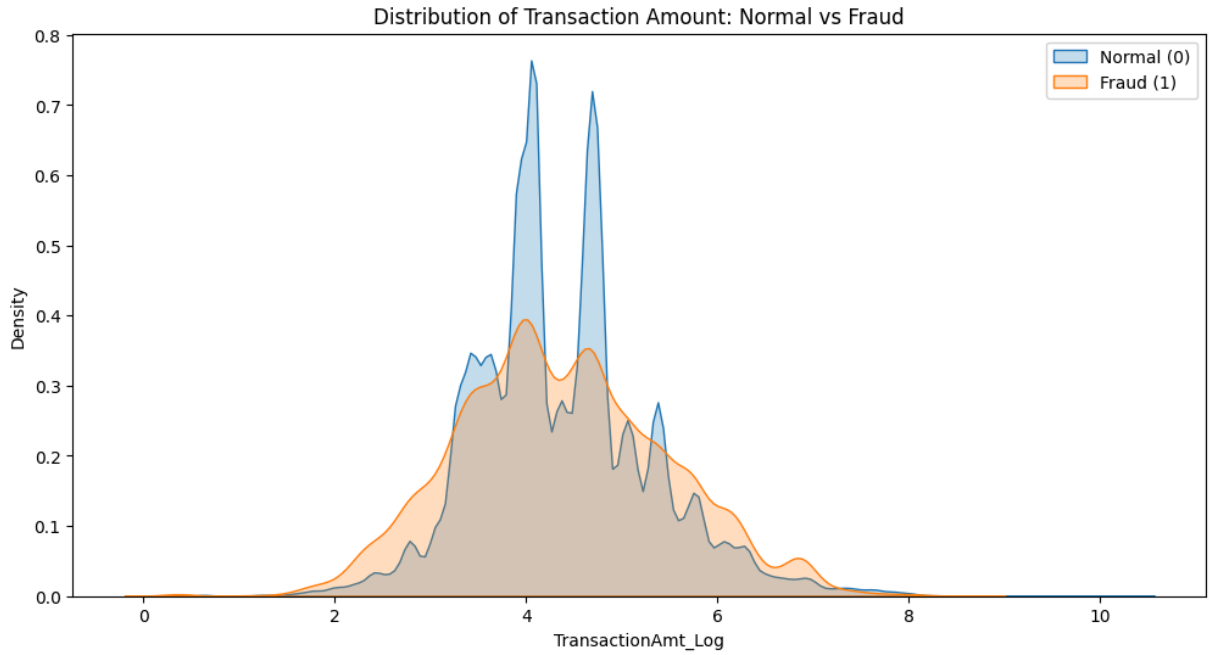
In [38]:
```
cat_cols_all = df_clean.select_dtypes(include=['object', 'category']).column
num_cols_all = df_clean.select_dtypes(include=['number', 'float16', 'float32
```

In [39]:
```
for col in cat_cols_all:
    if 'Unknown' not in df_clean[col].cat.categories:
        df_clean[col] = df_clean[col].cat.add_categories('Unknown')
    df_clean[col] = df_clean[col].fillna('Unknown')
df_clean[num_cols_all] = df_clean[num_cols_all].fillna(-1)
print("Total Missing in Full Data:", df_clean.isna().sum().sum())
```

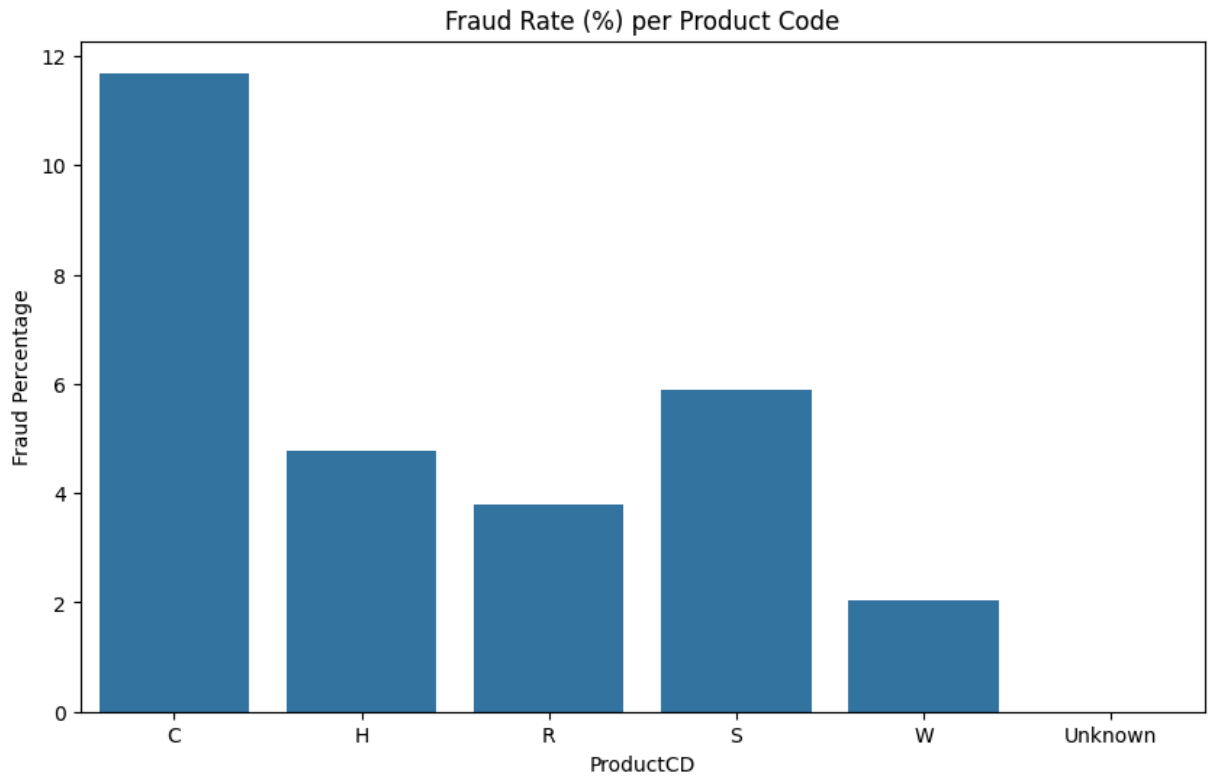
```
Total Missing in Full Data: 0
```

In [40]:
```
plt.figure(figsize=(12, 6))
sns.kdeplot(data=df_clean[df_clean['isFraud']==0]['TransactionAmt_Log'], lab
sns.kdeplot(data=df_clean[df_clean['isFraud']==1]['TransactionAmt_Log'], lab
plt.title('Distribution of Transaction Amount: Normal vs Fraud')
plt.legend()
plt.show()
```

```
/tmp/ipykernel_9564/217478267.py:2: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=df_clean[df_clean['isFraud']==0]['TransactionAmt_Log'], l
abel='Normal (0)', shade=True)
/tmp/ipykernel_9564/217478267.py:3: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=df_clean[df_clean['isFraud']==1]['TransactionAmt_Log'], l
abel='Fraud (1)', shade=True)
```



In [41]:
```
product_fraud = df_clean.groupby('ProductCD')['isFraud'].mean().sort_values(

plt.figure(figsize=(10,6))
sns.barplot(x=product_fraud.index, y=product_fraud.values)
plt.title('Fraud Rate (%) per Product Code')
plt.ylabel('Fraud Percentage')
plt.show()

print(product_fraud)
```




```
ProductCD
C    11.687269
S     5.899553
H     4.766231
R     3.782594
W     2.039939
Name: isFraud, dtype: float64
```

In [42]:
```
df_clean['Hour'] = (df_clean['TransactionDT'] / (3600)) % 24
df_clean['Hour'].head(10)
```

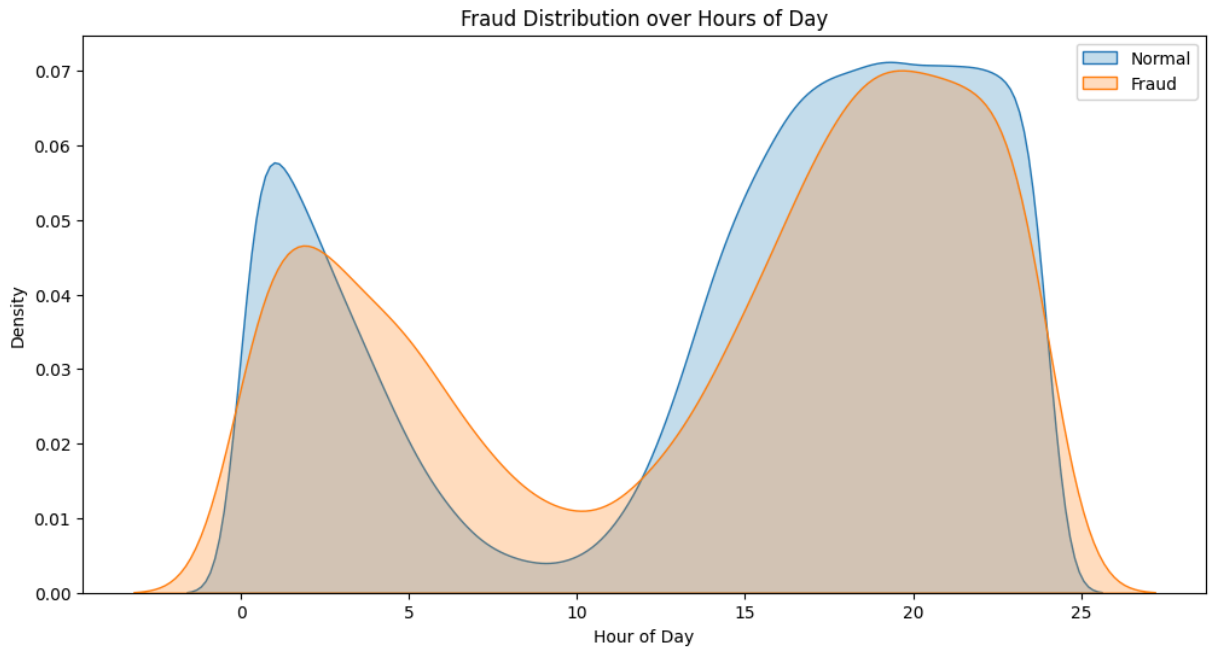
```
/tmp/ipykernel_9564/2163021260.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['Hour'] = (df_clean['TransactionDT'] / (3600)) % 24
```

Out[42]:
```
0    0.000000
1    0.000278
2    0.019167
3    0.027500
4    0.029444
5    0.030556
6    0.033889
7    0.035833
8    0.037500
9    0.037778
Name: Hour, dtype: float64
```

In [43]:
```
plt.figure(figsize=(12,6))
sns.kdeplot(data=df_clean[df_clean['isFraud']==0]['Hour'], label='Normal', s
sns.kdeplot(data=df_clean[df_clean['isFraud']==1]['Hour'], label='Fraud', sh
```

```
plt.title('Fraud Distribution over Hours of Day')
plt.xlabel('Hour of Day')
plt.legend()
plt.show()
```

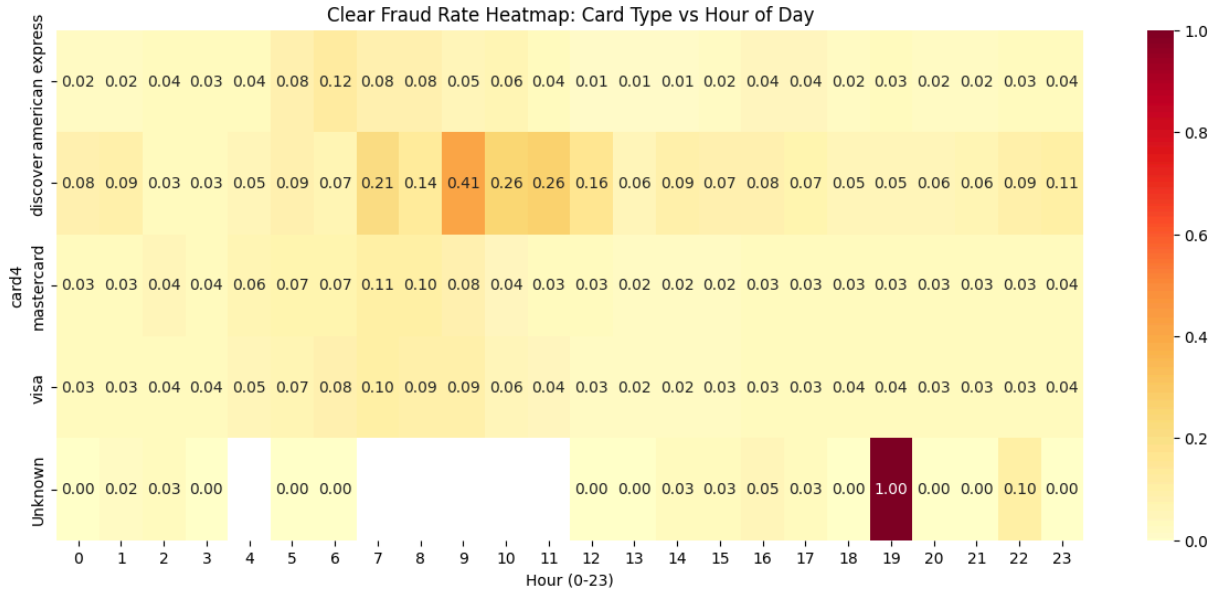
```
/tmp/ipykernel_9564/2095529546.py:2: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=df_clean[df_clean['isFraud']==0]['Hour'], label='Normal',
shade=True)
/tmp/ipykernel_9564/2095529546.py:3: FutureWarning:

`shade` is now deprecated in favor of `fill`; setting `fill=True`.
This will become an error in seaborn v0.14.0; please update your code.

  sns.kdeplot(data=df_clean[df_clean['isFraud']==1]['Hour'], label='Fraud',
shade=True)
```



In [44]:
```
df_clean['Hour_int'] = df_clean['Hour'].astype(int)
pivot = df_clean.pivot_table(values='isFraud', index='card4', columns='Hour_
```

```
/tmp/ipykernel_9564/4216436808.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['Hour_int'] = df_clean['Hour'].astype(int)
```

In [45]:
```
plt.figure(figsize=(15, 6))
sns.heatmap(pivot,annot=True,cmap='YlOrRd',fmt='.2f')
plt.title('Clear Fraud Rate Heatmap: Card Type vs Hour of Day')
plt.xlabel('Hour (0-23)')
plt.show()
```



In [46]:
```
df_clean.head()
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[46]:

	TransactionID	isFraud	TransactionDT	TransactionAmt	ProductCD	card1	card2	card3
0	2987000	0	86400	68.5	W	13926	-1.0	150.0
1	2987001	0	86401	29.0	W	2755	404.0	150.0
2	2987002	0	86469	59.0	W	4663	490.0	150.0
3	2987003	0	86499	50.0	W	18132	567.0	150.0
4	2987004	0	86506	50.0	H	4497	514.0	150.0

5 rows × 344 columns

In [47]:
```
df_clean['DeviceInfo'] = df_clean['DeviceInfo'].str.lower()
```

In [48]:
```
df_clean['device_combined'] = df_clean['DeviceType'].astype(str) + "_" + df_
print(df_clean['device_combined'].value_counts().head(10))
```

```
device_combined
Unknown_unknown                 449685
desktop_windows                  47720
mobile_ios device                19781
desktop_macos                    12565
desktop_unknown                  11715
mobile_unknown                   10474
desktop_trident/7.0               7407
desktop_rv:11.0                   1864
desktop_rv:57.0                    871
mobile_sm-j700m build/mmb29k       549
Name: count, dtype: int64
```

```
/tmp/ipykernel_9564/3935066362.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['device_combined'] = df_clean['DeviceType'].astype(str) + "_" + d
f_clean['DeviceInfo'].astype(str)
```

In [49]:
```
df_clean = df_clean.copy()
```

In [50]:
```
print(df_clean['addr2'].value_counts().head(5))
```

```
addr2
 87.0    520481
-1.0      65706
 60.0      3084
 96.0       638
 32.0        91
Name: count, dtype: int64
```

In [51]:
```
addr1_counts = df_clean['addr1'].value_counts()
df_clean['addr1_freq'] = df_clean['addr1'].astype('Float32').map(addr1_count
print(df_clean[['addr1', 'addr1_freq']].head())

addr1_counts = df_clean['addr1'].value_counts()
joblib.dump(addr1_counts, 'addr1_counts.pkl')
```

```
   addr1  addr1_freq
0  315.0       23078
1  325.0       42751
2  330.0       26287
3  476.0        9478
4  420.0        3581
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

Out[51]:
```
['addr1_counts.pkl']
```

In [53]:
```
print(df_clean['P_emaildomain'].value_counts().head(10))
```

```
P_emaildomain
gmail.com        228355
yahoo.com        100934
Unknown           94456
hotmail.com       45250
anonymous.com     36998
aol.com           28289
comcast.net        7888
icloud.com         6267
outlook.com        5096
msn.com            4092
Name: count, dtype: int64
```

In [54]:
```
df_clean['email_check'] = (df_clean['P_emaildomain'].astype('str') == df_cle

print(df_clean['email_check'].value_counts())
```

```
email_check
0    404644
1    185896
Name: count, dtype: int64
```

In [55]:
```
print(df_clean.groupby('email_check')['isFraud'].mean())
```

```
email_check
0    0.021337
1    0.064708
Name: isFraud, dtype: float64
```

In [56]:
```
df_clean['dist1_is_unknown'] = (df_clean['dist1'] == -1).astype(int)
df_clean['dist1_log'] = np.log1p(df_clean['dist1'].replace(-1, 0))
print(df_clean.groupby('dist1_is_unknown')['isFraud'].mean())
```

```
dist1_is_unknown
0    0.019956
1    0.045158
Name: isFraud, dtype: float64
```

In [57]:
```
scaler_d = MinMaxScaler()

d_nums = [1, 2, 3, 4, 5, 10, 11, 15]
d_original_cols = [f'D{i}' for i in d_nums]
d_log_cols = [f'D{i}_log' for i in d_nums]

df_clean['D1_is_unknown'] = (df_clean['D1'] == -1).astype(int)

for col in d_original_cols:
    df_clean[f'{col}_log'] = np.log1p(df_clean[col].clip(lower=0))

df_clean[d_log_cols] = scaler_d.fit_transform(df_clean[d_log_cols])

if 'D1' in df_clean.columns and 'D2' in df_clean.columns:
    df_clean['D1_minus_D2'] = df_clean['D1'] - df_clean['D2']

print("Done! D columns processed correctly.")
joblib.dump(scaler_d, 'scaler_d.pkl')
```

```
Done! D columns processed correctly.
```

Out[57]: ['scaler_d.pkl']

In [ ]:
```
print("Max values for D logs:")
print(df_clean[d_log_cols].max())

print("\nMin values for D logs:")
print(df_clean[d_log_cols].min())

if 'D1_minus_D2' in df_clean.columns:
    print("\nSample of D1_minus_D2 (First 5):")
    print(df_clean['D1_minus_D2'].head())
```

```
Max values for D logs:
D1_log     1.0
D2_log     1.0
D3_log     1.0
D4_log     1.0
D5_log     1.0
D10_log    1.0
D11_log    1.0
D15_log    1.0
dtype: float16

Min values for D logs:
D1_log     0.0
D2_log     0.0
D3_log     0.0
D4_log     0.0
D5_log     0.0
D10_log    0.0
D11_log    0.0
D15_log    0.0
dtype: float16

Sample of D1_minus_D2 (First 5):
0    15.0
1     1.0
2     1.0
3     0.0
4     1.0
Name: D1_minus_D2, dtype: float16
```

```
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
/home/mostafashazly/miniconda3/lib/python3.13/site-packages/pandas/io/format
s/format.py:1466: RuntimeWarning: overflow encountered in cast
  has_large_values = (abs_vals > 1e6).any()
```

In [58]:
```
c_cols = [f'C{i}' for i in range(1, 15) if f'C{i}' in df_clean.columns]

for col in c_cols:
    df_clean[f'{col}_is_unknown'] = (df_clean[col] == -1).astype(int)

scaler_c = MinMaxScaler()
```

```
for col in c_cols:
    df_clean[f'{col}_log'] = np.log1p(df_clean[col].clip(lower=0))

c_log_cols = [f'{col}_log' for col in c_cols]
df_clean[c_log_cols] = scaler_c.fit_transform(df_clean[c_log_cols])

print("Correlation with Fraud (C1_log):")
print(df_clean[['C1_log', 'isFraud']].corr()['isFraud'])
joblib.dump(scaler_c, 'scaler_c.pkl')
```

```
Correlation with Fraud (C1_log):
C1_log     0.08305
isFraud    1.00000
Name: isFraud, dtype: float64
```

Out[58]: ['scaler_c.pkl']

In [ ]:
```
m_cols = [f'M{i}' for i in range(1, 10) if f'M{i}' in df_clean.columns]

for col in m_cols:
    df_clean[col] = df_clean[col].astype(str).str.lower()

    df_clean[col] = df_clean[col].map({'t': 1, 'f': 0, 'true': 1, 'false': 0

print(f"Sample of M columns after encoding:")
print(df_clean[m_cols].head())
```

```
Sample of M columns after encoding:
    M1   M2   M3   M4   M5   M6   M7   M8   M9
0  1.0  1.0  1.0 -1.0  0.0  1.0 -1.0 -1.0 -1.0
1 -1.0 -1.0 -1.0 -1.0  1.0  1.0 -1.0 -1.0 -1.0
2  1.0  1.0  1.0 -1.0  0.0  0.0  0.0  0.0  0.0
3 -1.0 -1.0 -1.0 -1.0  1.0  0.0 -1.0 -1.0 -1.0
4 -1.0 -1.0 -1.0 -1.0 -1.0 -1.0 -1.0 -1.0 -1.0
```

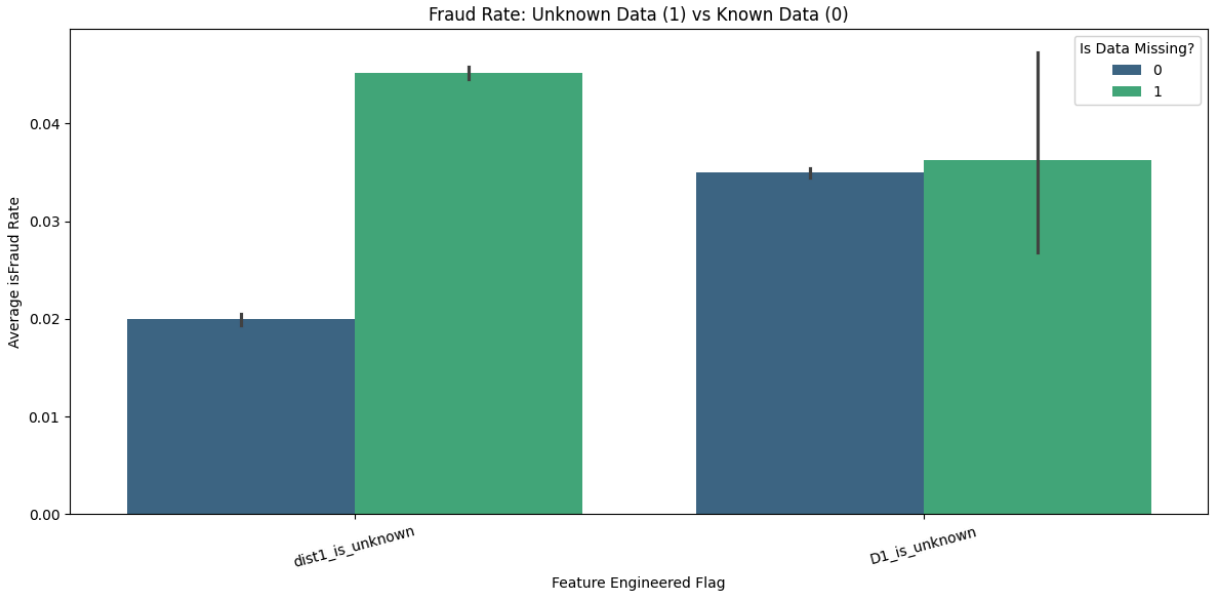
In [59]:
```
v_cols = [f'V{i}' for i in range(1, 340) if f'V{i}' in df_clean.columns]
scaler_v = MinMaxScaler()
df_clean[v_cols] = scaler_v.fit_transform(df_clean[v_cols].fillna(-1))
joblib.dump(scaler_v, 'scaler_v.pkl')
```

Out[59]: ['scaler_v.pkl']

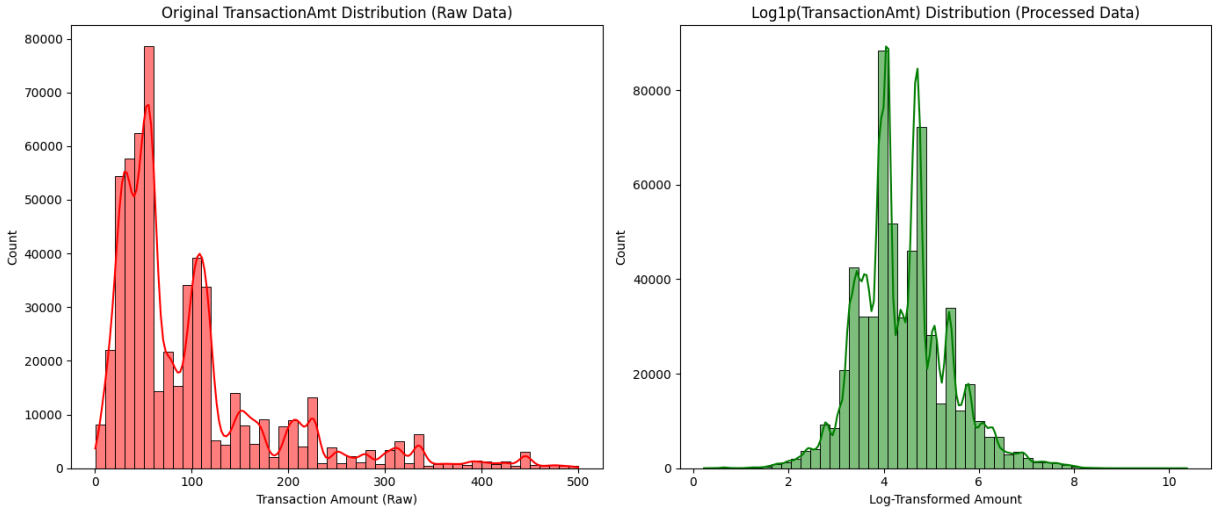
In [ ]:
```
df_clean['hour'] = (df_clean['TransactionDT'] // 3600) % 24
df_clean['day_of_week'] = (df_clean['TransactionDT'] // (3600 * 24)) % 7
```

```
/tmp/ipykernel_5556/406333078.py:1: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['hour'] = (df_clean['TransactionDT'] // 3600) % 24
/tmp/ipykernel_5556/406333078.py:2: PerformanceWarning: DataFrame is highly
fragmented.  This is usually the result of calling `frame.insert` many time
s, which has poor performance.  Consider joining all columns at once using p
d.concat(axis=1) instead. To get a de-fragmented frame, use `newframe = fram
e.copy()`
  df_clean['day_of_week'] = (df_clean['TransactionDT'] // (3600 * 24)) % 7
```

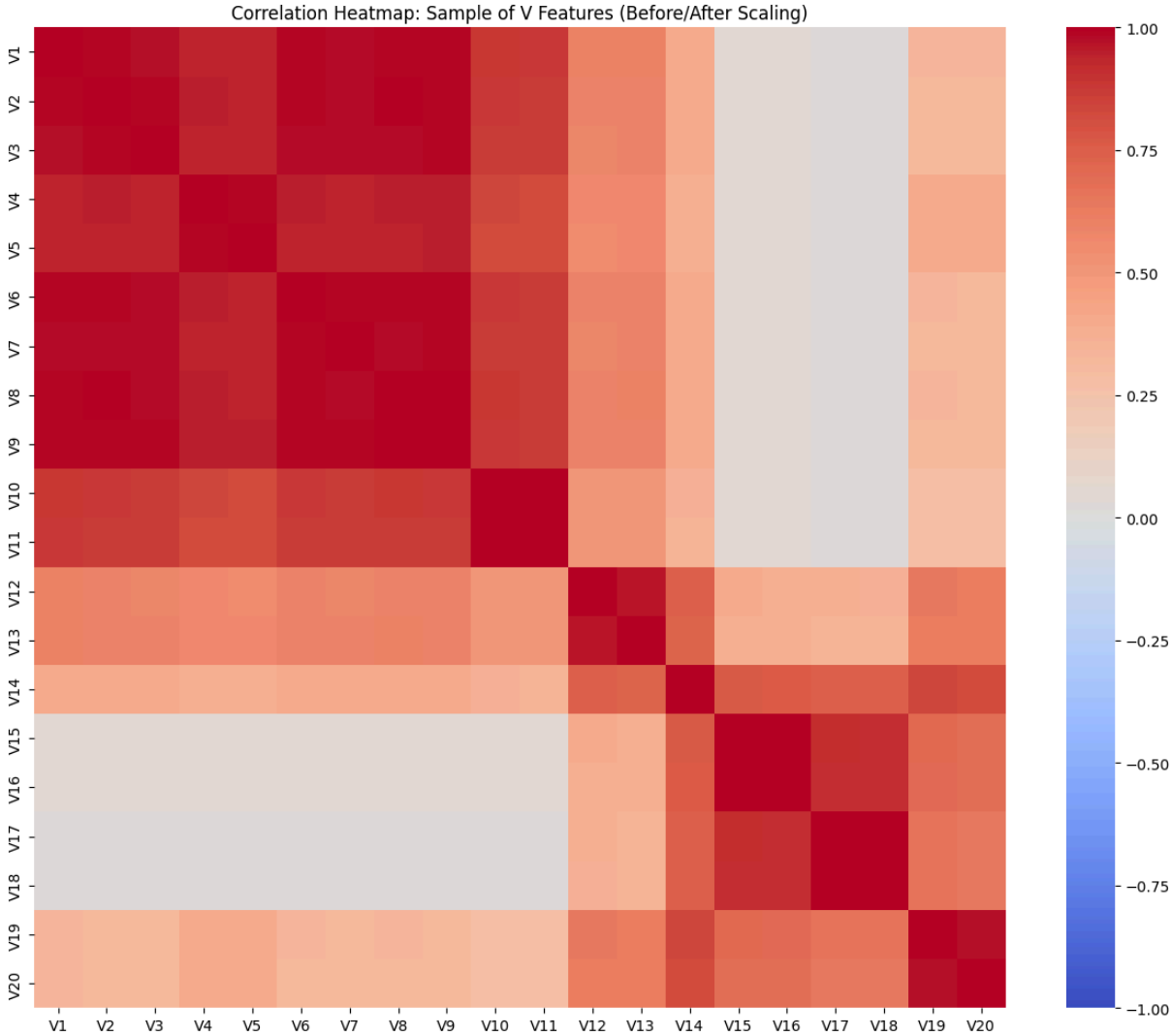
In [ ]:
```
flags = ['dist1_is_unknown', 'D1_is_unknown']
df_melted = df_clean.melt(id_vars=['isFraud'], value_vars=flags, var_name='F

plt.figure(figsize=(12, 6))
sns.barplot(data=df_melted, x='Flag_Type', y='isFraud', hue='Is_Mising', pal
plt.title('Fraud Rate: Unknown Data (1) vs Known Data (0)')
plt.ylabel('Average isFraud Rate')
plt.xlabel('Feature Engineered Flag')
plt.xticks(rotation=15)
plt.legend(title='Is Data Missing?')
plt.tight_layout()
plt.show()
```



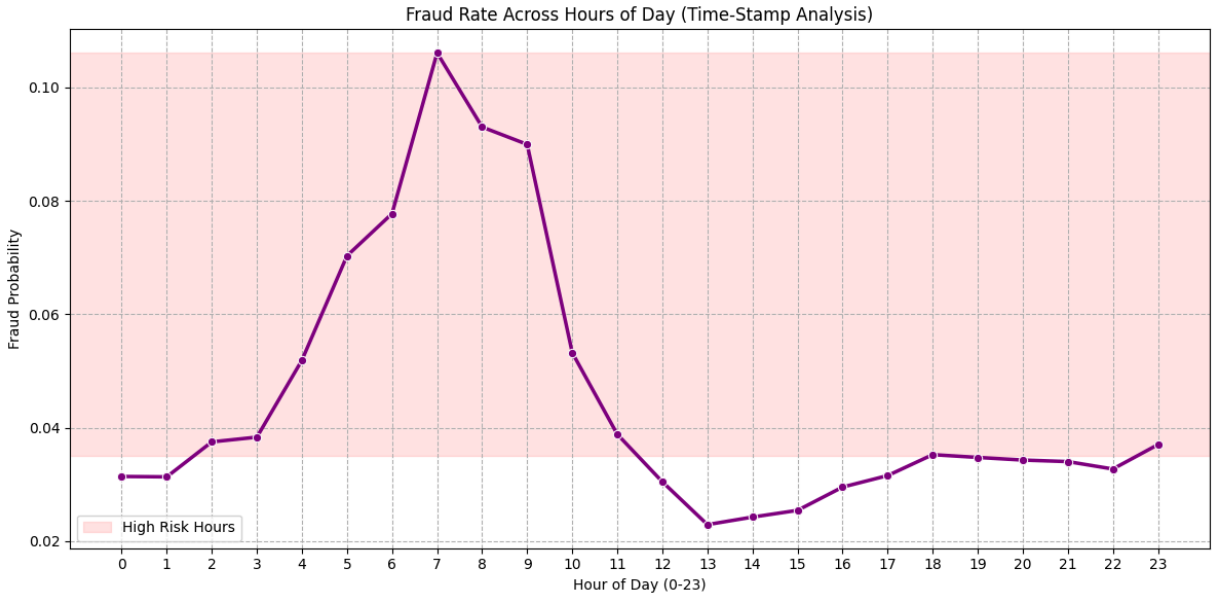
In [ ]:
```
plt.figure(figsize=(14, 6))

# 1. قبل الـ Log (Skewed Data)
plt.subplot(1, 2, 1)
sns.histplot(df_clean[df_clean['TransactionAmt'] < 500]['TransactionAmt'], b
plt.title('Original TransactionAmt Distribution (Raw Data)')
plt.xlabel('Transaction Amount (Raw)')

# 2. بعد الـ Log (Normalized Data)
plt.subplot(1, 2, 2)
# على العمود ده عشان تظهر المقارنة log هنفترض إننا لسه معملناش الـ
sns.histplot(np.log1p(df_clean['TransactionAmt']), bins=50, kde=True, color=
plt.title('Log1p(TransactionAmt) Distribution (Processed Data)')
plt.xlabel('Log-Transformed Amount')

plt.tight_layout()
plt.show()
```



In [ ]:
```
v_sample_cols = [f'V{i}' for i in range(1, 21) if f'V{i}' in df_clean.columr
corr_matrix = df_clean[v_sample_cols].corr()

plt.figure(figsize=(12, 10))
sns.heatmap(corr_matrix, cmap='coolwarm', annot=False, vmin=-1, vmax=1)
plt.title('Correlation Heatmap: Sample of V Features (Before/After Scaling)'
plt.tight_layout()
plt.show()
```



In [ ]:
```
df_clean['Hour_int'] = df_clean['Hour'].astype(int)

hourly_fraud = df_clean.groupby('Hour_int')['isFraud'].mean().reset_index()

plt.figure(figsize=(12, 6))
sns.lineplot(data=hourly_fraud, x='Hour_int', y='isFraud', marker='o', color

mean_fraud = df_clean['isFraud'].mean()
plt.axhspan(mean_fraud, hourly_fraud['isFraud'].max(), color='red', alpha=0.

plt.title('Fraud Rate Across Hours of Day (Time-Stamp Analysis)')
plt.ylabel('Fraud Probability')
plt.xlabel('Hour of Day (0-23)')
plt.xticks(range(24))
plt.grid(True, linestyle='--')
plt.legend()
plt.tight_layout()
plt.show()
```



In [ ]:
```
cols_to_drop = [
    'TransactionID', 'TransactionDT', 'TransactionAmt',
    'dist1', 'Hour', 'D1', 'D2', 'D3', 'D4', 'D5', 'D10', 'D11', 'D15']

c_original = [f'C{i}' for i in range(1, 15)]
cols_to_drop.extend(c_original)

df_final = df_clean.drop(columns=cols_to_drop)

print(f"Original columns: {len(df_clean.columns)}")
print(f"Final columns: {len(df_final.columns)}")
```

```
Original columns: 389
Final columns: 362
```

In [ ]:
```
df_final.columns.to_list()
```

Out[ ]:
```
['isFraud',
 'ProductCD',
 'card1',
 'card2',
 'card3',
 'card4',
 'card5',
 'card6',
 'addr1',
 'addr2',
 'P_emaildomain',
 'R_emaildomain',
 'M1',
 'M2',
 'M3',
 'M4',
 'M5',
 'M6',
 'M7',
 'M8',
 'M9',
 'V1',
 'V2',
 'V3',
 'V4',
 'V5',
 'V6',
 'V7',
 'V8',
 'V9',
 'V10',
 'V11',
 'V12',
 'V13',
 'V14',
 'V15',
 'V16',
 'V17',
 'V18',
 'V19',
 'V20',
 'V21',
 'V22',
 'V23',
 'V24',
 'V25',
 'V26',
 'V27',
 'V28',
 'V29',
 'V30',
 'V31',
 'V32',
 'V33',
 'V34',
 'V35',
```

```
 'C5_is_unknown',
 'C6_is_unknown',
 'C7_is_unknown',
 'C8_is_unknown',
 'C9_is_unknown',
 'C10_is_unknown',
 'C11_is_unknown',
 'C12_is_unknown',
 'C13_is_unknown',
 'C14_is_unknown',
 'C1_log',
 'C2_log',
 'C3_log',
 'C4_log',
 'C5_log',
 'C6_log',
 'C7_log',
 'C8_log',
 'C9_log',
 'C10_log',
 'C11_log',
 'C12_log',
 'C13_log',
 'C14_log',
 'hour',
 'day_of_week']
```

In [ ]:

```
dict_encoders = {}
cat_cols = df_final.select_dtypes(include=['object', 'category']).columns.to

for col in cat_cols:
    le = LabelEncoder()
    df_final[col] = le.fit_transform(df_final[col].astype(str))
    dict_encoders[col] = le

joblib.dump(dict_encoders, 'label_encoders_dict.pkl')

X = df_final.drop(columns=['isFraud'])
y = df_final['isFraud']

scaler = MinMaxScaler()
X_scaled = pd.DataFrame(scaler.fit_transform(X), columns=X.columns)

joblib.dump(scaler, 'min_max_scaler.pkl')

df_final_ready = pd.concat([X_scaled, y.reset_index(drop=True)], axis=1)
df_final_ready.to_csv('train_final_processed.csv', index=False)
```